

PRODUCT REQUIREMENTS DOCUMENT

StrategyFlow AI Agent

From Visual Strategy Builder to Autonomous Financial AI Agent

STATUS	VERSION	LAST UPDATED	PRIORITY
Discovery	1.0	2025	Critical

1. Executive Summary

StrategyFlow AI Agent transforms an existing visual node-and-edge strategy builder into a fully autonomous, 24/7 financial AI agent. Rather than serving as a passive flowchart tool that users reference manually, the upgraded platform acts as a proactive AI coworker — monitoring markets, reading financial news and reports, executing trades, managing portfolio risk, and alerting users in real time.

The product democratizes capabilities that have historically been reserved for institutional hedge funds — systematic strategy execution, around-the-clock market surveillance, news-aware portfolio management, and automated trade execution — and makes them accessible to any individual investor, active trader, or wealth manager.

The core innovation is treating the user's flowchart not as a diagram, but as a living, executable decision engine that an AI agent traverses in real time as market conditions evolve.

2. Problem Statement

2.1 The Gap in Existing Tools

Current financial tools fall into two unsatisfying categories:

- Passive dashboards and charting tools that require the user to constantly monitor screens and manually execute decisions based on their strategy.
- Robo-advisors (Betterment, Wealthfront) that offer generic, static rebalancing with no ability to incorporate custom user logic, real-time news, or nuanced market conditions.
- Traditional trading bots that react only to price action, with no ability to interpret context (e.g., understanding why a stock is dropping — market correction vs. CEO resignation).

2.2 The Core Pain

Users build sophisticated mental strategies but cannot execute them consistently because:

- Markets move 24/7 (especially crypto) but humans need to sleep.
 - Relevant news breaks at any hour and requires immediate portfolio assessment.
 - Emotional decisions override well-designed strategic logic under pressure.
 - The cognitive load of monitoring multiple assets, reading reports, and tracking news simultaneously is impossible for a single person.
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3. Product Vision & Goals

3.1 Vision Statement

"Give every person access to their own AI-powered hedge fund team — one that never sleeps, reads everything, knows their exact strategy and portfolio, and acts on their behalf the moment an opportunity or risk appears."

3.2 Success Metrics

Metric	Target (6 months)	Stretch Goal (12 months)
Agent Uptime	99.5%	99.9%
Signal-to-Alert Latency	< 60 seconds	< 10 seconds
Strategy Execution Accuracy	95% match to flowchart logic	99% match
User Retention (90-day)	40%	65%
Autonomous Actions per User/Month	50+	200+

4. Target Users & Real-World Personas

Persona 1 — The Retail Investor (Primary)

Background	Employed full-time. Invests in stocks and ETFs with a long-term strategy. Can't watch markets during work hours.
Pain Point	Has a solid strategy on paper but breaks it emotionally when markets move. Misses buying opportunities while at work.
Job To Be Done	"Execute my strategy consistently without me having to babysit the market."
How They Use It	Builds a flowchart: 'If quality stock drops 10% with no fundamental news, buy \$500.' Receives a WhatsApp alert when this triggers with option to approve or let agent auto-execute.
Replaces	Personal financial advisor, robo-advisor

Persona 2 — The Active Trader (Primary)

Background	Day trader or swing trader who monitors multiple tickers and strategies simultaneously.
Pain Point	Cannot scan 50 tickers at once, read every earnings report, and track sentiment — cognitive overload.
Job To Be Done	"Be my second brain — filter out the noise and alert me only when my exact setup is met."
How They Use It	Builds complex multi-condition nodes: technical triggers (RSI, volume), news filters, sector momentum. Agent scans the entire market and alerts with a pre-filled trade recommendation.
Replaces	Junior analyst, trading alert services (TradingView alerts, Trade Ideas)

Persona 3 — The Wealth Manager / RIA (Secondary)

Background	Registered Investment Advisor managing 20–100 client portfolios, each with different risk profiles and strategies.
Pain Point	Cannot monitor all client portfolios in real time. Quarterly reviews miss time-sensitive opportunities and risks.
Job To Be Done	"Scale my business by letting AI monitor all client portfolios continuously so I only step in when something important happens."

How They Use It	Creates a separate strategy flowchart per client. Agent monitors all portfolios simultaneously and surfaces a prioritized alert dashboard for the advisor to review and act on.
Replaces	Associate analysts, manual portfolio review process

Persona 4 — The Crypto Trader (Secondary)

Background	Active crypto investor. Markets never close, creating constant anxiety about missing moves during sleep.
Pain Point	3am crashes and pumps happen regularly. No way to manage risk around the clock.
Job To Be Done	"Protect my portfolio and capture opportunities even when I'm asleep."
How They Use It	DCA strategy with take-profit and stop-loss rules triggered by on-chain data, fear/greed index, and sentiment. Agent executes autonomously 24/7.
Replaces	Crypto trading bots (3Commas, Pionex) but with news-awareness and natural language strategy definition

5. Market Disruption Map

StrategyFlow AI Agent does not sit in a vacuum. It directly disrupts a set of expensive, inaccessible, or inefficient real-world financial roles — and augments a set of professionals who currently rely on slow, manual workflows. This section maps exactly where the product sits in the financial services landscape.

5.1 — Roles This Tool Replaces (For Retail Users)

The following roles are typically only accessible to high-net-worth individuals or institutional investors. StrategyFlow makes them available to anyone with a strategy and an internet connection.

Role Replaced	What They Do (That We Automate)	Our Advantage
Hedge Fund Team	Portfolio manager + quant analyst + risk manager + news analyst working together as a team. Costs \$500k+/year in salaries.	Full team capability delivered by one AI agent. Available to any individual investor at a subscription price.
Stockbroker	Calls clients with opportunities, executes trades, monitors positions, collects commission on every trade.	24/7 autonomous monitoring with zero commission bias. Executes only what the user's strategy dictates.
Financial Advisor	Reviews portfolio quarterly, recommends rebalancing, checks if strategy still makes sense given market conditions.	Continuous review every heartbeat cycle — not quarterly. Always aware of breaking news and macro shifts.
Quant Trader	Writes algorithmic code to systematically execute rules-based strategies. Requires deep programming skills.	No-code visual builder replaces the programming requirement. Strategy flowchart IS the algorithm.
Junior Analyst	Reads FOMC reports, earnings calls, and financial filings. Synthesizes data into memos for senior staff.	Agent reads and synthesizes in milliseconds. Never fatigued, never misses a filing, never needs sleep.
Robo-Advisor (Betterment, Wealthfront)	Offers static, generic rebalancing based on basic risk profile. No custom logic, no news-awareness.	Dynamic, news-aware, fully personalised strategy execution. User defines every rule — not a generic algorithm.
Static Trading Bots (3Commas, Pionex)	React only to price action via hard-coded rules. Cannot interpret context or reason about why a stock is moving.	Interprets context: understands WHY a stock is dropping (CEO resigned vs. normal correction) before acting.

5.2 — Roles This Tool Augments (For Professionals)

For professionals who already hold these roles, StrategyFlow acts as a force-multiplier — letting them scale their practice without adding headcount, and making them faster and more responsive than competitors.

Professional	How They Use StrategyFlow	The Force-Multiplier Effect
Independent Trader / Day Trader	Builds multi-condition strategy nodes (RSI + volume + sentiment + news filters). Agent monitors 50+ tickers simultaneously so they only step in when the perfect setup is confirmed.	One trader effectively monitors the whole market. No more missed setups because they were watching a different ticker.
RIA / Boutique Wealth Manager	Creates a separate strategy flowchart per client. Agent monitors all portfolios simultaneously and surfaces a prioritized morning alert dashboard.	Manages 3-5x more clients without adding junior staff. Competitive edge over firms using quarterly manual reviews.
Family Office	Uses the agent to monitor a diversified multi-asset portfolio across stocks, bonds, real estate funds, and crypto with a single unified strategy layer.	Institutional-grade monitoring at a fraction of the cost of building an in-house quant team.
Retail Investor (Active)	Builds their strategy once in the visual builder. Agent executes it consistently — without emotional deviation — while they are at work or asleep.	Removes the #1 retail investing failure: abandoning a good strategy due to emotional decisions under market pressure.
Crypto Trader	Sets DCA rules, take-profit nodes, and stop-loss triggers. Agent runs 24/7 since crypto markets never close — including 3am pumps and crashes.	Full market coverage without needing to be awake. No missed opportunities, no panic sells while sleeping.

5.3 — The Big Picture: Democratizing Institutional Finance

The tools hedge funds use — systematic strategies, 24/7 monitoring, automated execution, news analysis, risk management — have historically only been available to people with millions to invest and entire teams to operate them. StrategyFlow makes all of that available to anyone with a strategy and an internet connection.

Think of it this way: imagine if every person could have their own Bloomberg terminal, their own quant analyst, and their own portfolio manager — all inside one app, all running on their personal strategy, all autonomous. That is the final goal.

Previously Only For Hedge funds & institutions	StrategyFlow Delivers To every individual investor	At a Fraction of the Cost \$29–\$499/month vs. \$500k+/year
• 24/7 market monitoring • Systematic strategy execution • Real-time news analysis • Automated trade execution • Portfolio risk management	• Retail investors with a strategy • Active traders & day traders • Independent wealth managers • Crypto investors (24/7 markets) • Family offices without quant teams	• Starter: \$29/month • Pro: \$99/month • Wealth Mgr: \$499/month • vs. \$500k+/year for a hedge fund team

6. Feature Requirements

Priority Legend: P0 = Must Have (MVP) | P1 = Should Have | P2 = Nice to Have. Effort: S = Small | M = Medium | L = Large | XL = Extra Large

6.1 — Autonomous Heartbeat Engine

Background Agent Daemon (Heartbeat Loop)

P0

L

A persistent background process that wakes the agent at configurable intervals, independent of the UI. Based on OpenClaw's heartbeat architecture. The agent runs continuously whether the user is logged in or not.

- Configurable polling intervals (e.g., every 5 mins, on market open/close, on trigger events)
- Agent runs as a background worker process (BullMQ / Celery queue) decoupled from the web server
- Per-user agent instances that load the user's active strategy flowchart on each cycle
- Heartbeat health monitoring with auto-restart on failure
- Agent state persistence — remembers context between heartbeat cycles

6.2 — Strategy Flowchart Execution Engine

Node-to-Logic Compiler

P0

XL

The existing visual flowchart is converted into a machine-executable decision tree. The agent traverses node conditions in real time as market data comes in, evaluating each branch against live data feeds.

- Parse node/edge graph into an Abstract Syntax Tree (AST) or directed acyclic graph
- Support condition nodes (IF/AND/OR logic), action nodes (buy/sell/alert), and data source nodes
- Real-time evaluation of conditions against live market data streams
- Branch probability scoring — agent can rank which branch paths are closest to triggering
- Strategy simulation / backtesting mode to validate logic before going live

6.3 — Market Data & Intelligence

Multi-Source Data Ingestion

P0

L

The agent is equipped with tool-calling capabilities to autonomously fetch and synthesize data from multiple financial data sources in real time.

- Real-time price, volume, and order book data via Polygon.io or Alpaca Markets API

- Technical indicator calculation (RSI, MACD, Bollinger Bands, moving averages)
- Earnings calendar and economic event schedule awareness
- Options flow and unusual activity monitoring
- Crypto on-chain data integration (Glassnode, Santiment)

News & Sentiment Analysis

P1

L

The agent autonomously reads and interprets financial news, social sentiment, and regulatory announcements — then maps findings to the user's portfolio and active strategy nodes.

- Real-time news ingestion via NewsAPI, Bloomberg (if licensed), or RSS aggregators
- SEC EDGAR filings monitoring (10-K, 10-Q, 8-K event-driven alerts)
- Social sentiment aggregation: Reddit (WallStreetBets), Twitter/X FinTwit, StockTwits
- LLM-powered news-to-portfolio impact analysis — 'Does this news affect my positions?'
- Sentiment scoring with confidence levels fed into condition nodes as inputs

6.4 — Proactive Multi-Channel Notifications

Outbound Agent Communications

P0

M

When the agent identifies an opportunity or risk that meets the user's defined thresholds, it initiates contact — the user does not need to check the app. The agent reaches out.

- WhatsApp and SMS alerts via Twilio API with contextual, plain-English explanations
- Voice call capability for critical alerts (e.g., major drawdown, margin call risk)
- Push notifications via Firebase for mobile app users
- Email summaries for non-urgent periodic reports (daily/weekly digest)
- Alert includes: what triggered, why it matters, what action the agent recommends or took
- Human-in-the-Loop mode: agent texts user awaiting 'YES/NO' reply before executing

6.5 — Autonomous Trade Execution

Brokerage API Integration

P1

XL

The agent can execute trades and manage portfolio positions autonomously via connected brokerage accounts, subject to user-defined guardrails established in the strategy flowchart.

- Alpaca Markets integration for stocks and crypto (free paper trading for testing)
- Interactive Brokers (IBKR) integration for professional users
- Robinhood API integration for retail user base
- Position sizing logic — agent respects max allocation per trade defined in strategy nodes

- Guardrail nodes: max daily spend limits, drawdown stop thresholds, position concentration caps
- Paper trading mode — full simulation before live funds are committed
- Full audit log of every agent action with rationale for compliance and trust

6.6 — Portfolio Awareness

Portfolio Context Engine	P0	M
All agent analysis and decisions are contextualized against the user's real current portfolio. The agent never makes generic recommendations — every signal is filtered through the lens of what the user actually holds. • Real-time portfolio sync from connected brokerage accounts • Position tracking: cost basis, unrealized P&L, sector exposure, concentration risk • Correlation analysis — agent understands how holdings relate to each other • Tax-awareness layer: tracks short vs. long-term gains to inform sell decisions • Portfolio health score updated on each heartbeat cycle		

7. High-Level System Architecture

7.1 Core Layers

Layer	Components & Technologies
Presentation Layer	Existing React node/edge flowchart builder UI + new Agent Dashboard
Strategy Compiler	Graph parser (flowchart to AST to decision tree). TypeScript or Python. Compiles at strategy save time.
Agent Orchestrator	LangGraph or custom tool-calling loop. Manages agent state, tool dispatch, and decision routing.
Heartbeat Scheduler	BullMQ (Node.js) or Celery (Python). Job queue triggers per-user agent instances on defined intervals.
Tool Layer	Function-calling tools: fetch_price_data, read_news, analyze_sentiment, execute_trade, send_alert
Data Sources	Polygon.io, Alpha Vantage, SEC EDGAR, NewsAPI, Reddit API, Twitter/X API, Glassnode (crypto)
Execution Layer	Alpaca API, IBKR API, Robinhood API. Wrapped in guardrail middleware that validates against strategy limits.
Notification Layer	Twilio (SMS/WhatsApp/Voice), Firebase (push), SendGrid (email)
Memory & State	Postgres for structured data. Pinecone/Weaviate vector store for agent memory, news summaries, past analysis.
Agent Memory	Short-term: current heartbeat context. Long-term: vector store of past signals, user behavior, portfolio history.

7.2 Operational Modes

Mode	Description	Best For
Advisory Mode	Agent monitors and alerts but never executes. User approves all actions via reply or in-app.	New users, risk-averse investors
Human-in-the-Loop	Agent executes automatically but sends a pre-action notification with a cancel window (e.g., 2 min to cancel).	Active traders who want speed but oversight
Fully Autonomous	Agent executes all trades defined in the strategy flowchart without confirmation. Full guardrails apply.	Experienced users with backtested strategies
Simulation Mode	Agent runs all logic against live data but makes no real trades or alerts. Used for strategy testing.	Strategy development and backtesting

8. Key Use Cases & User Stories

8.1 The Sleep-Well Risk Protector

User Story: As a retail investor, I want my agent to automatically protect my portfolio from major losses overnight so that I wake up knowing my risk parameters were respected while I slept.

Strategy Flowchart Logic: Monitor tech holdings → IF NASDAQ drops >3% overnight OR negative regulatory AI news detected → liquidate 30% of tech holdings to cash → send WhatsApp message with action summary.

Agent Action at 3:00 AM: Heartbeat cycle detects NASDAQ futures down 3.2%. News tool surfaces new EU AI regulation announcement. Condition nodes evaluate to TRUE. Agent executes partial liquidation via Alpaca API. Twilio sends: 'Emergency protocol triggered. Moved \$10,200 to cash to protect your portfolio. EU AI regulation news triggered your risk rule.'

8.2 The Alpha Hunter

User Story: As an active trader, I want my agent to scan the entire market 24/7 for setups that match my exact criteria so that I never miss a trade because I wasn't watching.

Strategy Flowchart Logic: Scan all S&P 500 tickers → IF RSI < 35 AND volume > 2x 20-day average AND sentiment score > 0.7 positive AND no negative earnings in last 30 days → alert user with pre-filled trade recommendation.

Agent Action: During market hours, agent detects NVDA matching all conditions following an analyst upgrade. Agent sends alert: 'NVDA matches your breakout setup. RSI: 32. Volume: 3.1x average. Sentiment: 0.82 positive. No negative news. Recommend \$2,000 position. Reply YES to execute.'

8.3 The Wealth Manager Multiplier

User Story: As an RIA, I want to monitor all 40 of my client portfolios simultaneously so that I can spot risks and opportunities across my entire book of business without adding headcount.

Agent Action: Each client has a unique strategy flowchart. The agent runs separate heartbeat instances per client portfolio. The advisor receives a morning digest: '3 portfolios require attention today: Client A — earnings risk on 2 positions. Client B — rebalancing threshold triggered. Client C — stop-loss approaching on TSLA.' Advisor reviews and approves recommended actions from a single dashboard.

9. Product Roadmap

Phase	Name	Deliverables	Timeline
1	Foundation — The Always-On Agent	- Background heartbeat scheduler (BullMQ) - Flowchart-to-decision-tree compiler (AST parser) - Real-time market data integration (Polygon / Alpaca) - Advisory mode (monitor + alert, no trade execution) - WhatsApp / SMS alert system (Twilio)	Weeks 1–6
2	Intelligence Layer	- News ingestion + LLM sentiment analysis - SEC EDGAR filing monitoring - Portfolio sync from brokerage (read-only) - Portfolio health scoring on each heartbeat - Agent memory via vector store (past analysis context)	Weeks 7–12
3	Execution & Autonomy	- Paper trading mode with Alpaca sandbox - Live trade execution with guardrail middleware - Human-in-the-loop approval flow (2-min cancel window) - Fully autonomous mode with strategy-defined limits - Position sizing and risk management nodes	Weeks 13–20
4	Scale & Monetization	- Multi-portfolio support (wealth manager persona) - Backtesting engine against historical data - Strategy marketplace (users share flowchart templates) - Voice call alerts for critical events - Mobile app with agent dashboard	Weeks 21–30

10. Non-Functional Requirements

9.1 Security & Compliance

- All brokerage API keys encrypted at rest using AES-256. Never stored in plaintext.
- OAuth 2.0 for brokerage account connections — never store user credentials directly.
- Full audit log of every agent decision and action for regulatory compliance.
- Users must explicitly grant execution permissions — default is advisory mode only.
- SEC/FINRA regulatory disclosure: platform is not a registered investment advisor. All autonomous actions are user-configured rules, not financial advice.

9.2 Reliability & Performance

- Agent heartbeat uptime target: 99.5% (Phase 1), 99.9% (Phase 3+).
- Signal-to-alert latency: < 60 seconds from market event to user notification.
- Graceful degradation: if a data source is unavailable, agent logs the gap and continues with available data rather than stopping.
- Auto-restart on agent failure with exponential backoff — user notified if agent is offline for > 15 minutes.

9.3 Guardrails (Non-Negotiable)

- Maximum single-trade value cap (user-defined, enforced by middleware — cannot be overridden by agent).
- Maximum daily spend limit across all autonomous actions.
- Portfolio concentration limit: agent cannot allocate more than X% to any single position.
- Market circuit breaker: agent pauses all autonomous actions during circuit breaker events.
- Emergency kill switch: user can halt all agent activity instantly via app, SMS reply, or web dashboard.

11. Monetization Strategy

Tier	Features	Price	Target User
Free	Visual strategy builder only. No agent. No autonomous actions. Up to 3 strategies.	Free	Hobbyist, evaluation

Starter	Agent monitoring + alerts. Advisory mode only. 1 connected portfolio. 5 strategies.	\$29 / month	Retail investor
Pro	Full autonomous execution. Unlimited strategies. Multi-asset. Priority alerts.	\$99 / month	Active trader
Wealth Manager	Multi-client portfolios. Team access. White-label option. Compliance exports.	\$499 / month	RIA / family office

12. Open Questions & Risks

#	Risk / Open Question	Mitigation / Decision Needed
1	Regulatory risk: does autonomous trade execution require the platform to register as a broker-dealer or RIA?	Legal review required. Default advisory mode only until legal structure is confirmed. Consider 'rules engine' framing, not 'investment advice.'
2	LLM hallucination in financial decisions — agent acts on incorrect news interpretation.	Confidence thresholds required before action. Human-in-the-loop default. Dual-source news verification before triggering critical actions.
3	Brokerage API rate limits may prevent real-time execution at scale.	Alpaca has generous limits. Implement request queuing. Evaluate co-location options for Pro/Enterprise tiers.
4	User trust: will users actually grant execution permissions to an AI agent?	Start with advisory mode. Build trust with accuracy track record. Show simulation results before live deployment. Transparent audit log is essential.
5	Data source cost at scale: Polygon, Bloomberg, and Twitter APIs are expensive.	Start with free tiers (Alpaca, Yahoo Finance, Reddit). Gate premium data sources behind paid tiers. Model unit economics at 1k, 10k, 100k users.

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Questions? Reach your product team to schedule a strategy review session.